IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.0

Project ID: R26-ISE-002

1. Topic (12 words max)

Rack-Level Micro-Climate and Growth Optimizer for Oyster Mushroom Farms

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering (ISE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (500 words max) – references not included in word count.

Oyster mushroom cultivation is an increasingly important agricultural activity in Sri Lanka due to its low initial investment, short cultivation cycle, and growing market demand. However, maintaining optimal growth conditions throughout the mushroom life cycle remains a major challenge for farmers. Oyster mushrooms are highly sensitive to environmental factors, and these requirements vary significantly across different biological growth stages including early growth, pinning, fruiting, and maturity. In farms that use multi-rack shelving systems, environmental conditions are not uniformly distributed across the cultivation space. Each rack and even each shelf can experience a different micro-climate due to variations.

Among these factors, the lack of accurate biological stage awareness combined with the absence of rack-level environmental visibility is a major limitation in current cultivation practices. Farmers typically do not have a reliable mechanism to determine the exact growth stage of mushrooms and instead depend on visual judgment and experience. Consequently, environmental controls such as mist sprayers and exhaust fans are operated using fixed schedules or trial-and-error methods, without considering the biological stage or localized micro-climate conditions. Furthermore, moisture management is often based solely on relative humidity values, even though the same humidity level can produce significantly different physiological effects at different temperatures due to variations in transpiration rates. As a result of this combined lack of stage awareness and temperature-humidity interpretation, farmers frequently encounter cap drying, pin abortion, uneven fruit body development, and yield inconsistency, even when recommended humidity levels are seemingly maintained.

In addition, improper moisture management contribute to the accumulation of carbon dioxide within cultivation racks, leading to localized physiological stress and disorders that are not immediately visible. Existing CO_2 monitoring approaches, where available, mostly operate independently of biological growth context, limiting their effectiveness in early stress prevention.

A further critical challenge is disease identification. Oyster mushroom farmers in Sri Lanka still rely heavily on manual visual inspection to identify diseases appearing on mushroom bags and fruiting bodies. This process is slow, subjective, and often inaccurate, especially for early-stage infections that are difficult to notice with the naked eye. As a result, diseases such as green mold and black mold can spread rapidly across racks, causing significant yield loss and compromising product quality. Although prior studies have explored mushroom disease detection, many fail to reflect the environmental and operational conditions specific to Sri Lankan oyster mushroom farms. Furthermore, farmers typically lack accessible digital tools that can analyze images and provide immediate, rack-level alerts about contamination.

Therefore, the core research problem addressed by this project is the absence of an integrated, intelligent, rack-level micro-climate and growth management system that understands biological growth stage, models effective moisture stress using a temperature-humidity based transpiration stress index, predicts physiological stress using CO_2 trend analysis with hybrid image-based validation, and detects major oyster mushroom diseases early through image-based identification.

Addressing this problem directly contributes to zero hunger (SDG 2), decent work and economic growth (SDG 8) and responsible consumption and production (SDG 12).

References:

https://www.researchgate.net/publication/390771402_Enhancing_Oyster_Mushroom_Cultivation_with_Solar-Powered_IoT_and_Machine_Learning_Predicting_Harvest_Readiness

https://www.researchgate.net/publication/350570529_Design_of_temperature_and_humidity_control_system_on_oyster_mushroom_plant_house_based_on_Internet_of_Things_IoT

https://www.mdpi.com/2071-1050/13/10/5434

https://www.researchgate.net/publication/352787330_Major_Fungal_Contaminants_of_Mushrooms_and_Their_Management

6. Brief description of the nature of the solution including a conceptual diagram (500 words max)

The proposed solution is an integrated, rack-level intelligent micro-climate and growth optimization system delivered through a mobile application and a portable sensing-and-actuation device, designed to support oyster mushroom cultivation in small and medium scale farms.

The camera mounted on the portable device captures images of fruiting bodies. These images are transmitted to the mobile application and backend system, where computer vision techniques are used to identify the current biological growth stage, such as early growth, pinning, fruiting, or maturity. Visual features are analyzed. The detected stage provides essential biological context and is clearly displayed in the mobile application. The system can also estimate the remaining time until harvest based on detected growth characteristics.

Using the identified biological growth stage as contextual input, the moisture management module evaluates effective moisture stress at rack level. Rack-level temperature and relative humidity data collected from sensors integrated into the portable device are jointly analyzed to model temperature-humidity interaction. A temperature-humidity based transpiration stress index is computed and based on the calculated stress level and the detected growth stage, a rule-based decision logic is applied to determine appropriate corrective actions. Using this decision logic, the system automatically activates the mist sprayer to increase humidity or the exhaust fan to reduce humidity.

In parallel, a CO_2-based physiological disorder detection module continuously monitors carbon dioxide concentration using the CO_2 sensor embedded in the same portable device. A normalized CO_2 stress index is computed using time-series analysis to detect rising or prolonged stress conditions. When abnormal CO_2 levels persist beyond defined thresholds, the mobile application generates early warnings and recommends corrective actions such as increased ventilation. If stress continues, the system uses the same camera in the portable device to capture images for image-based validation, confirming physiological disorders.

Additionally, an image-based disease detection module operates, where farmers scan mushroom bags or fruiting bodies. The captured images are transmitted to the backend system for automated analysis, enabling the identification of common fungal infections such as green mold and black mold. Detected infections are visually localized within the images to clearly indicate affected regions. Farmers can further confirm and record the health status of each rack as healthy or infected within the application, creating a rack-level reference history that supports disease tracking over time.

The proposed solution operates using an input-process-output model, where rack-level environmental data such as temperature, humidity and CO_2 readings are collected from sensors integrated into the portable device, together with image inputs captured via the portable device camera and the mobile application, and farmer-confirmed rack health status. These inputs are processed to compute a transpiration stress index and a CO_2 stress index, while image data are analyzed using computer vision models to detect biological growth stages, identify fungal diseases, and validate physiological disorders when stress conditions persist. The system outputs actionable information through the mobile application, including detected growth stage, estimated time to harvest, rack-wise humidity and CO_2 status, stress and disease alerts, visually annotated images highlighting infected regions, and recommended corrective actions such as activating the mist sprayer or exhaust fan, while allowing farmers to record rack health status.

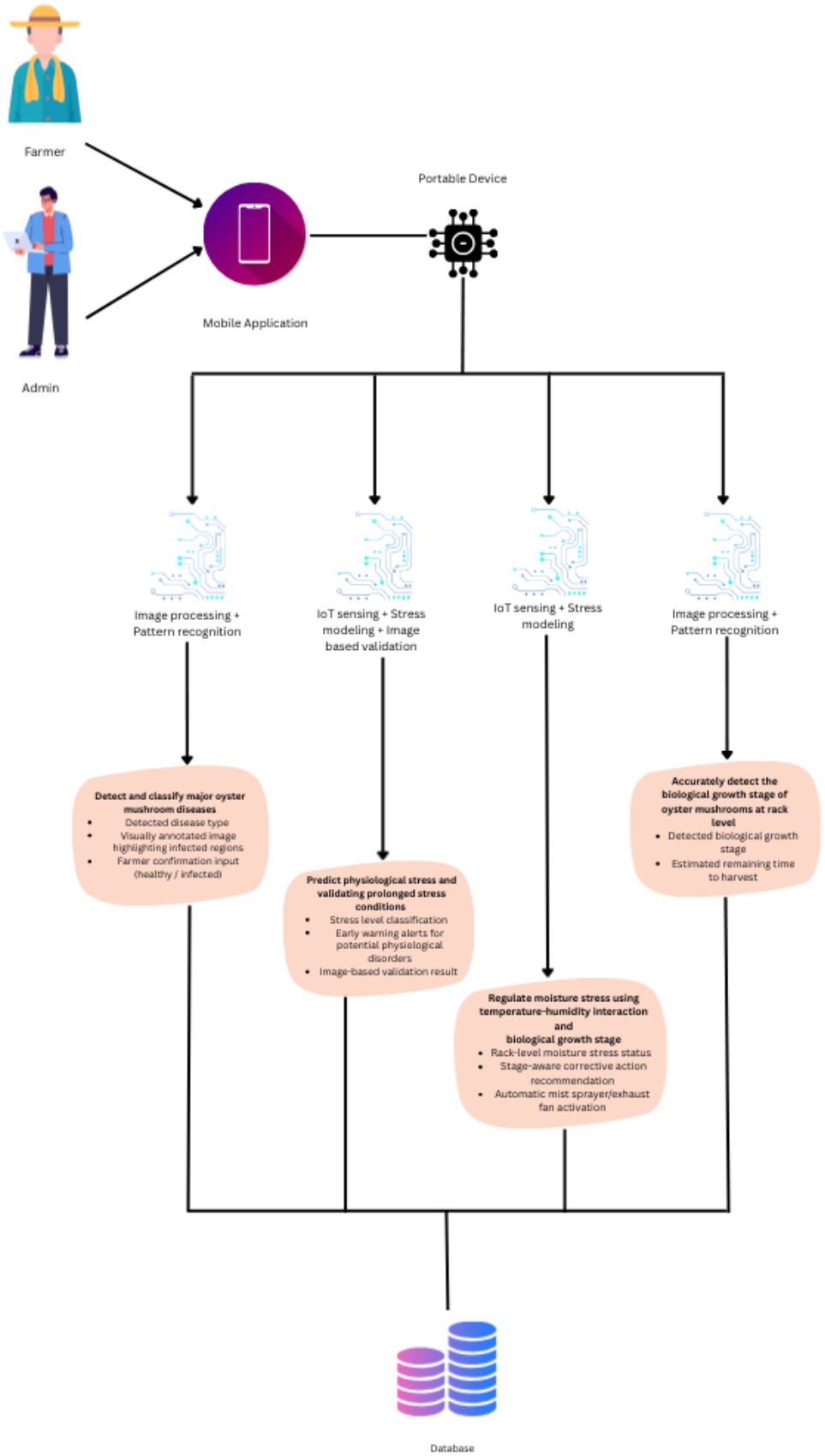

Farmer
Admin
Mobile Application
Portable Device
Image processing + Pattern recognition
IoT sensing + Stress modeling + Image based validation
IoT sensing + Stress modeling
Image processing + Pattern recognition
Detect and classify major oyster mushroom diseases
Detected disease type
Visually annotated image highlighting infected regions
Farmer confirmation input (healthy / infected)
Predict physiological stress and validating prolonged stress conditions
Stress level classification
Early warning alerts for potential physiological disorders
Image-based validation result
Regulate moisture stress using temperature-humidity interaction and biological growth stage
Rack-level moisture stress status
Stage-aware corrective action recommendation
Automatic mist sprayer/exhaust fan activation
Accurately detect the biological growth stage of oyster mushrooms at rack level
Detected biological growth stage
Estimated remaining time to harvest
Database

Figure 1 Conceptual Diagram

7. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

From a domain expertise perspective, we build on established knowledge in oyster mushroom cultivation, including biological growth stages, environmental requirements, moisture regulation, common physiological disorders, and fungal diseases. This domain understanding is supported through academic literature, agricultural extension resources, and insights obtained from local oyster mushroom farmers and cultivation centers in Sri Lanka, ensuring that the research remains practically relevant and context aware. Through discussions with experienced oyster mushroom farmers, agricultural officers, and cultivation experts, we gain firsthand knowledge of rack-level environmental behavior and biological growth management. Field visits to operational oyster mushroom grow rooms allow us to directly get practical exposure.

In terms of technical and specialized knowledge, image-based components apply computer vision techniques for biological stage detection and disease identification, while environmental monitoring components use data analytics, time-series analysis, and rule-based decision logic. Our technical skills are developed through structured learning using academic resources, online platforms, and academic supervision.

Regarding data availability, we utilize multiple data sources. Image data for biological stage detection and disease identification are collected directly from oyster mushroom farms using smartphones under varying lighting and growth conditions. Environmental data such as temperature, humidity and CO_2 concentration are obtained using commercially available sensors (e.g., DHT-series sensors for temperature and humidity and MQ-135 or MH-Z19 sensors for CO_2), installed at rack level to capture micro-climate variations. These sensors provide continuous, real-time readings suitable for analysis and model development. We manually annotate collected images by labeling growth stages, healthy samples, and disease-affected regions to support supervised learning. Data validations are supported through collaborations and access to recognized agricultural and research centers, including the Agriculture Research Centers at Gannoruwa, Makandura and Dambadeniya, and Industrial Technology Institute at Colombo.

All data collection is limited to environmental sensor readings and plant-based imagery. Nevertheless, we follow ethical research practices by obtaining permission from farm owners, avoiding the disclosure of identifiable farm information, and using collected data strictly for academic and research purposes.

8. Objectives and Novelty

Main Objective To design and develop an intelligent, rack-level micro-climate and growth optimization system for oyster mushroom cultivation that integrates image-based biological growth stage detection, temperature-humidity based transpiration stress modeling for moisture management, CO_2 driven physiological stress prediction, and image-based disease identification to improve yield consistency, reduce crop loss, and support small and medium-scale mushroom farmers.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
H. P. D. M Chandrasiri – IT22180148	Accurately detect the biological growth stage of oyster mushrooms at rack level using image analysis	• Capture images using a camera fixed to a portable device • Preprocess images for lighting and noise variations • Train a model to classify growth stages (early, pinning, fruiting, mature) • Display detected stage and remaining harvest days in the application	Introduces rack-level biological stage awareness instead of farmer estimation, converts visual growth characteristics into actionable biological data
R. T. S. Dissanayake – IT22062406	Regulate moisture stress at rack level using temperature-humidity interaction and	• Collect rack-level temperature and relative humidity data • Compute a temperature-humidity based transpiration stress index • Integrate detected biological growth stage into moisture stress evaluation	Introduces TSI-based moisture stress modeling that captures temperature-humidity interaction and biological context, moving beyond static humidity thresholds toward biologically meaningful micro-climate control.

	biological growth stage	• Apply rule-based decision logic to activate mist sprayers or exhaust fans based on TSI	
S. G. A. Sarathchandra – IT22182746	Predict CO_2-induced physiological stress and validate prolonged stress conditions using image-based confirmation	• Collect CO_2 readings continuously at rack level • Perform trend analysis to identify early stress conditions • Trigger early warnings when CO_2 stress exceeds defined thresholds • Automatically initiate image capture for affected racks if high CO_2 persists • Analyze captured images to confirm physiological disorders • Generate rack-wise corrective action recommendations	Provides rack-level, predictive CO_2 stress detection with hybrid image validation to deliver early, farmer-friendly warnings
M. I. Amana – IT22156624	Detect and classify major oyster mushroom diseases using image-based techniques	• Collect and annotate images of healthy and infected mushroom bags • Train a model to identify green mold and black mold • Localize infected regions visually • Allow farmers to confirm and record rack health status	Automated image-based disease detection tailored to Sri Lankan farm conditions, reduces reliance on subjective manual inspection

9. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

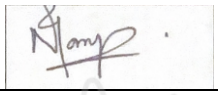
Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor	Ms.	Narmada	Gamage	
Co-Supervisor	Ms.	Rangi	Liyanage	
External Supervisor	-	-	-	-
Summary of external supervisor's (if any) experience and expertise				